Network Flow I
(ADA 2023, CSE 222)

Diptapriyo Majumdar

April 3, 2023

Outline

1. Network Flow

2. Max-Flow Min-Cut Theorem

Capacity (c) $c : E(G) \to \mathbb{R}^+$

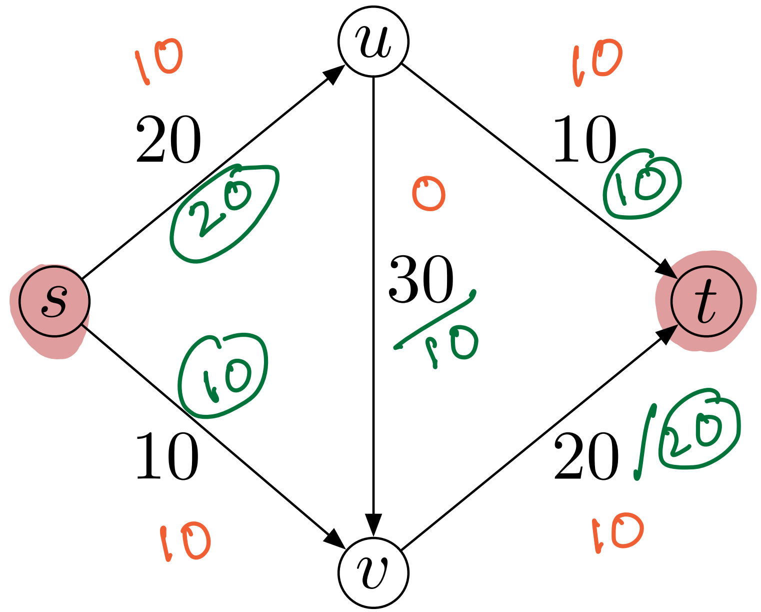
$(s,u) - 20$

$(s,v) - 10$

$(u,v) - 30$

$(u,t) - 10$

$(v,t) - 20$



flow (f)

$f : E(G) \to \mathbb{R}^+$

$(s,u) - 10/20$

$(u,t) - 10/10$

$(s,v) - 10/10$

$(v,t) - 10/20$

$(u,v) - 0/10$

Capacity Condition: $f(e) \le c(e)$

Flow Conservation: $\displaystyle\sum_{e \text{ into } u} f(e) = \sum_{e \text{ out of } u} f(e)$ for every u, except s and t.

- **Flow Network:** A directed graph $G = (V, E)$ with $s, t \in V(G)$ (s is source and t is sink). Every edge $e \in E(G)$ has a nonnegative *capacity* $\mathrm{c} : E(G) \to \mathbb{Z}^+$.
- **Defining flow:** What is *flow* or traffic in a network? An s-t *flow* in a network is a function $\mathrm{f} : E(G) \to \mathbb{Z}^+$ satisfying the following two conditions.
  - **Capacity condition:** For every $e \in E(G)$, $\mathrm{f}(e) \leq \mathrm{c}(e)$.
  - **Flow conservation condition:** For each u other than s and t, we have
    $$\sum_{v:(v,u) \in E(G)} \mathrm{f}(v, u) = \sum_{v:(u,v) \in E(G)} \mathrm{f}(u, v)$$

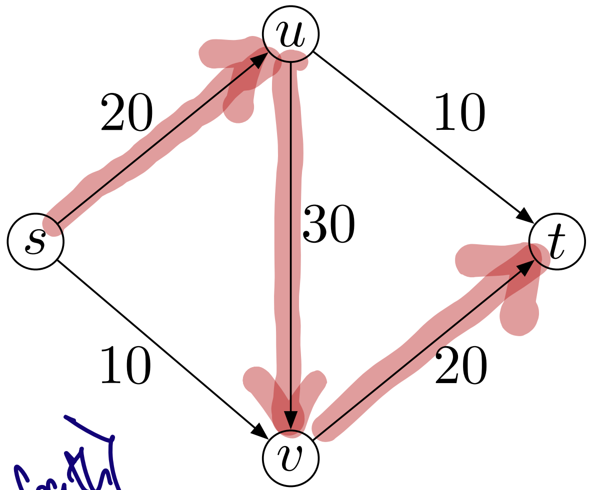
    More specifically $\mathrm{f}^{\mathrm{out}}(u) = \sum_{v:(u,v) \in E(G)} \mathrm{f}(u, v) = \sum_{e \text{ out of } u} \mathrm{f}(e)$ and $\mathrm{f}^{\mathrm{in}}(u) = \sum_{v:(v,u) \in E(G)} \mathrm{f}(v, u) = \sum_{e \text{ into } u} \mathrm{f}(e)$.
- **Input:** A flow network with capacity conditions flow conservation conditions.

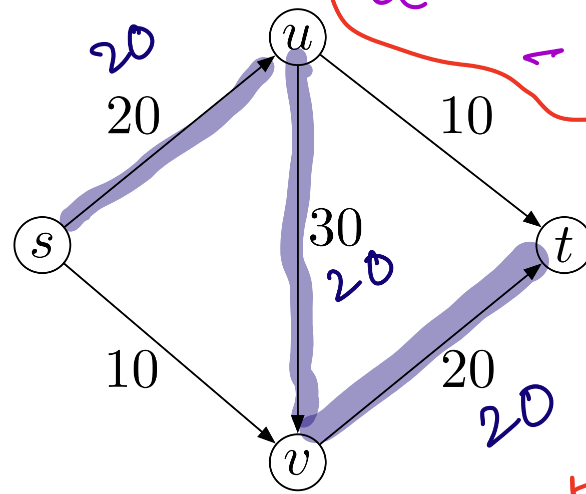
  (s,t) flow
- **Output:** A flow $\mathrm{f} : E(G) \to \mathbb{Z}^+$ that satisfies capacity conditions and flow conservation conditions, and maximize $\sum_{e \text{ out of } s} \mathrm{f}(e) = \sum_{u:(s,u) \in E(G)} \mathrm{f}(s, u)$.

$s \to u \to v \to t$

Bottleneck capacity of an (s,t)-path

$\min\{c(e) \mid e \in (s,t)\text{-path}\}$



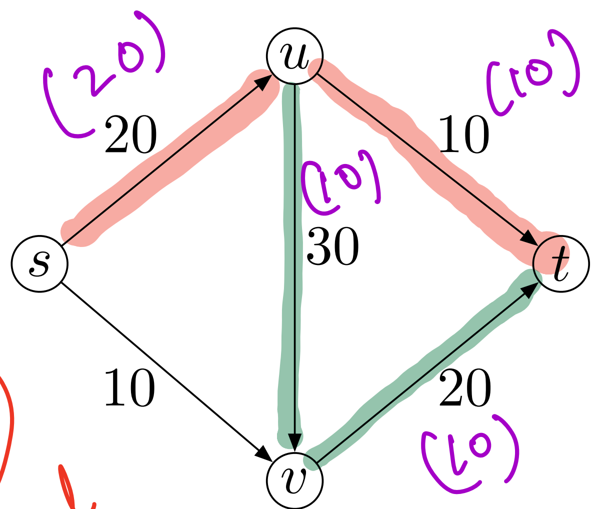


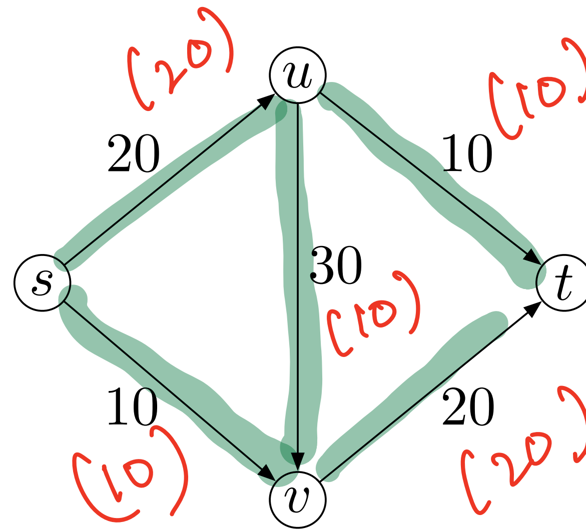
$c(u,v) - f(u,v) = 30 - 10 = 20$

residual capacity

If $f(u,v) = c(u,v)$

then (u,v) is saturated.

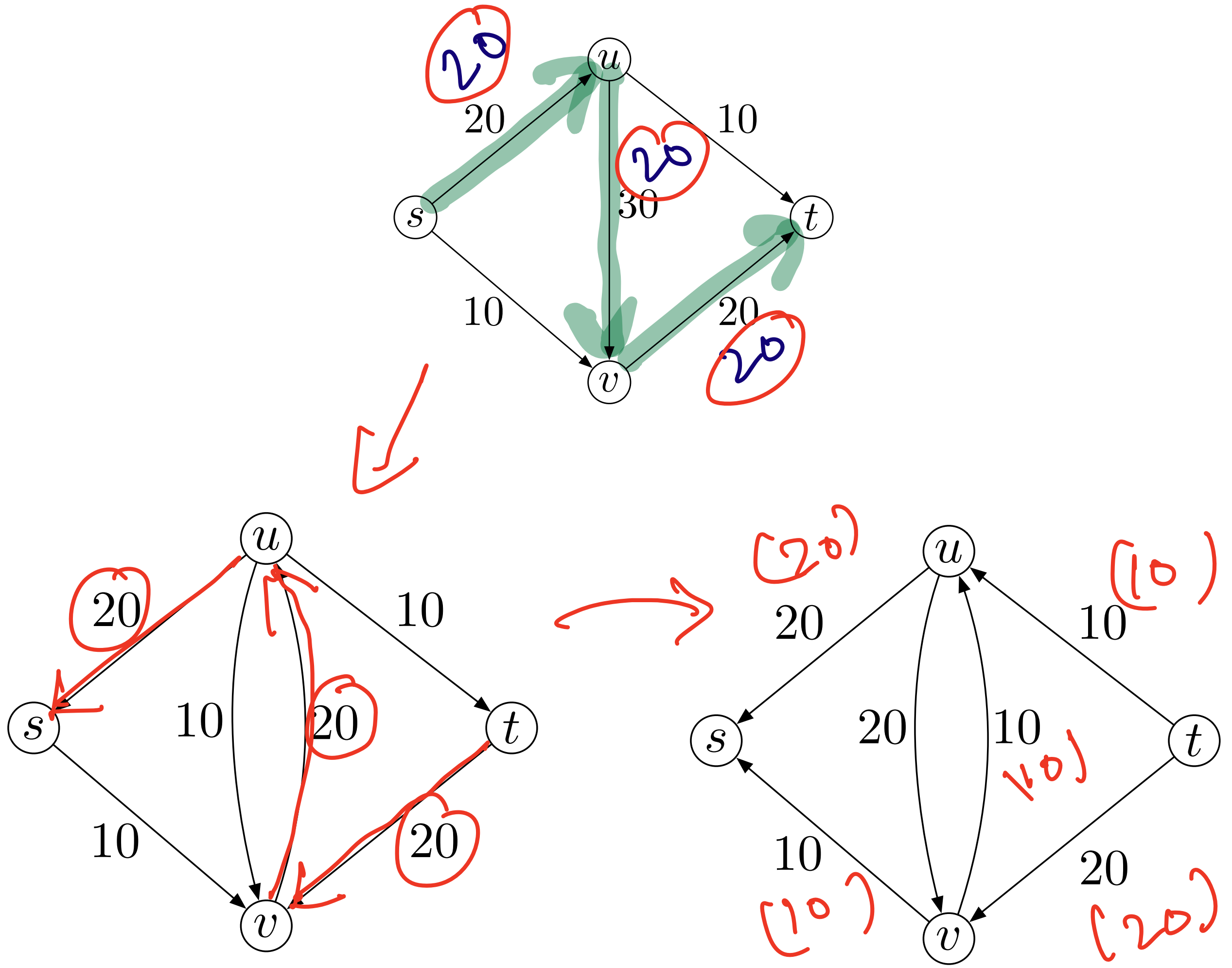




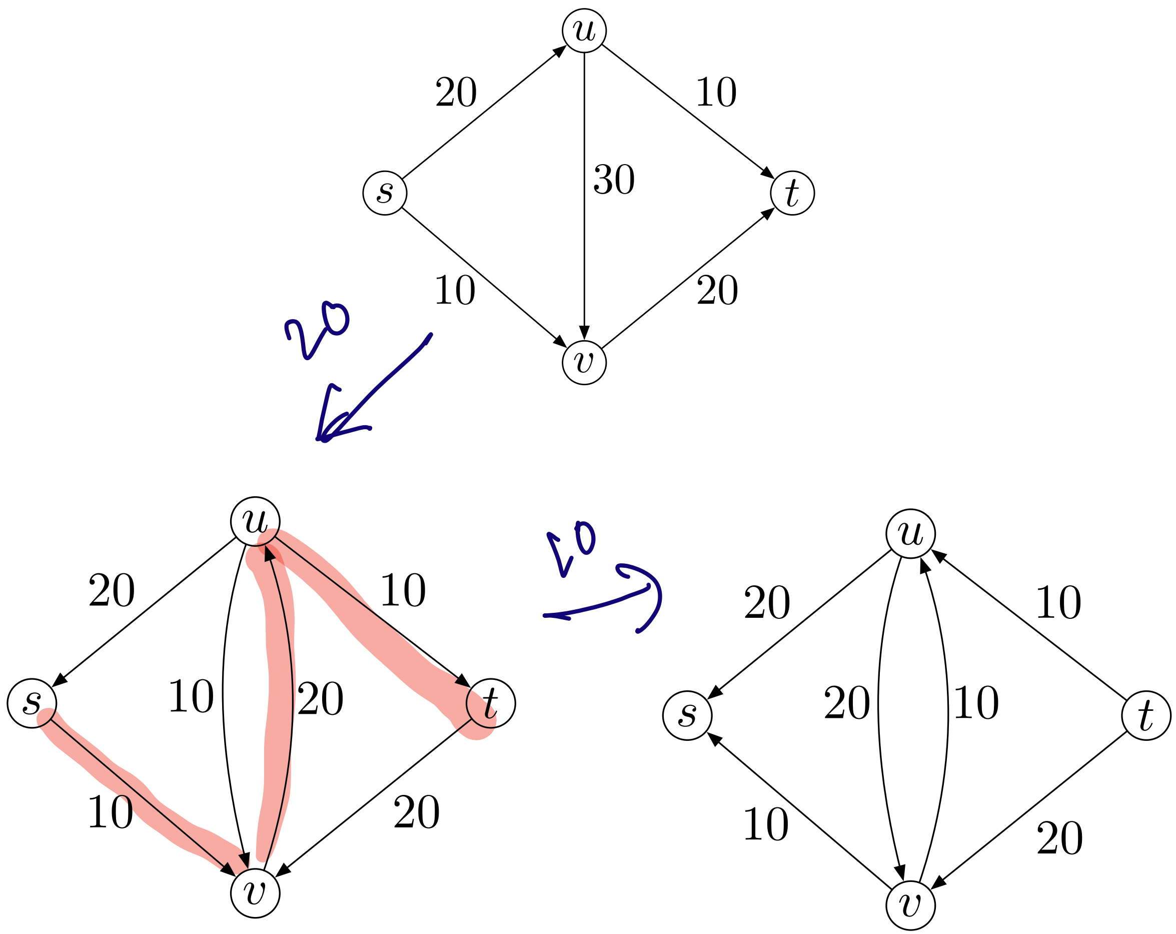
$c(v,t) - f(v,t) = 20 - 10 = 10$

residual capacity

- **Residual Graph:** Given a flow network G and a flow f on G, the *residual graph* G_f of G with respect to f as follows.



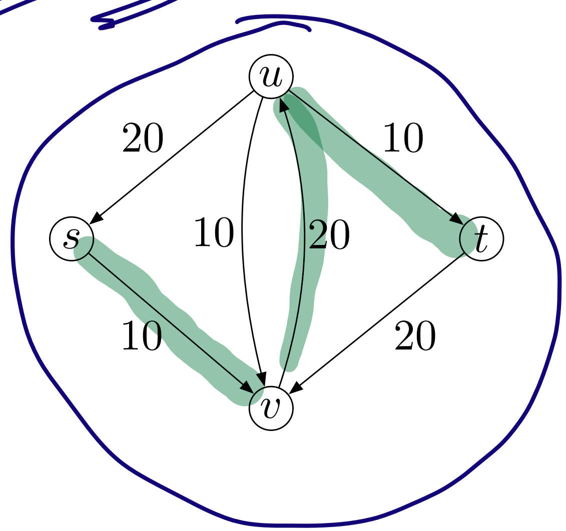
- **Augmenting Path:** An s-t path in residual graph is called *augmenting path*.

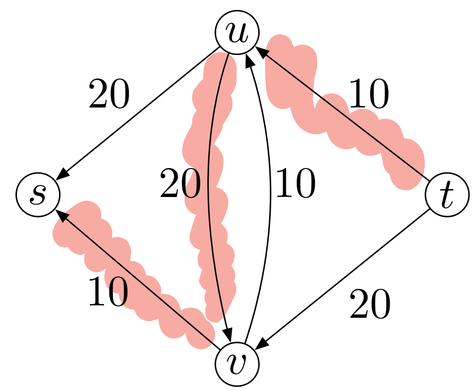


- Let G be a flow network and a f be a flow in G. Then, G_{f} has same vertex set as G.

- For every edge e, if $0 < \mathrm{f}(e) < \mathrm{c}(e)$, then $\cancel{\mathrm{f}(e) - \mathrm{c}(e)}$ $\mathrm{c}(e) - \mathrm{f}(e)$ is called *residual capacity*. By this forward edge e, this residual amount of flow $\mathrm{c}(e) - \mathrm{f}(e)$ can still be sent.

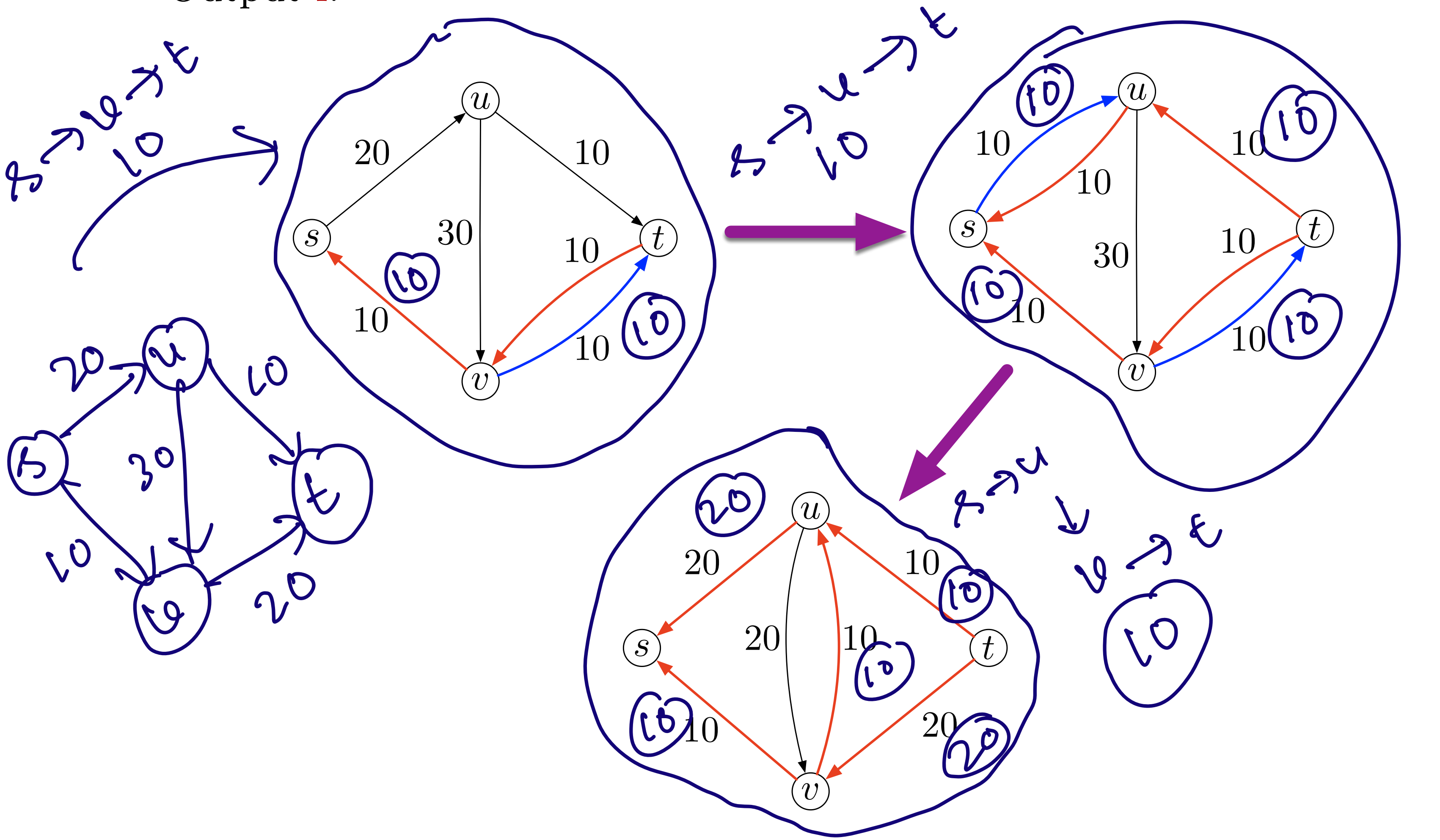
- For every edge e, if $0 < \mathrm{f}(e) < \mathrm{c}(e)$, then a backward edge e^{rev} is created with $\mathrm{c}(e^{rev}) = \mathrm{f}(e)$ in G_{f}.

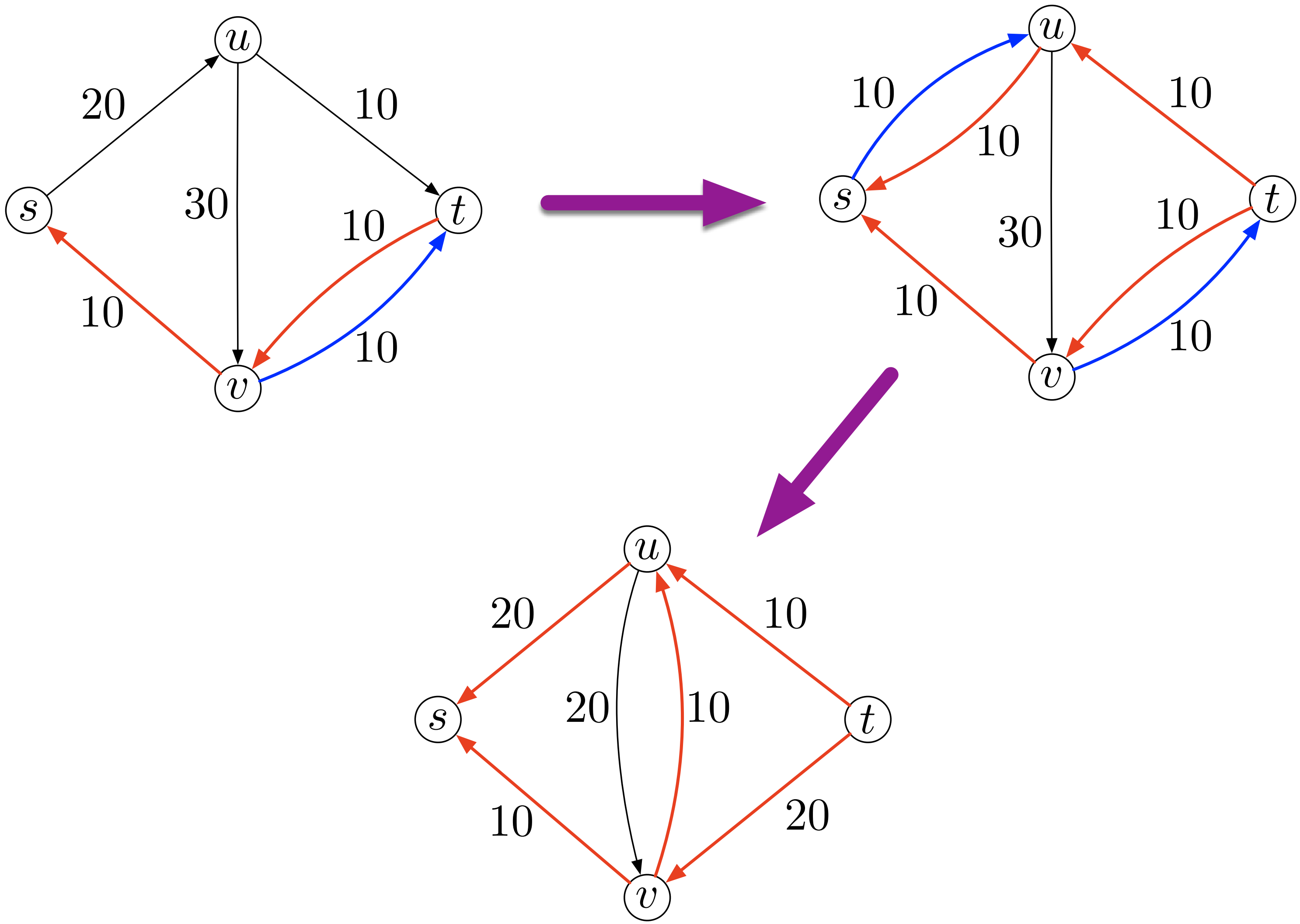




- For all $e \in E(G)$, initialize $f(e) = 0$ and $G_f = G$.
- As long as there is an augmenting path P in G_f
  - Send the maximum amount of flow via the path P from s to t.
  - Compute G_f.
- Output f.

bottleneck capacity

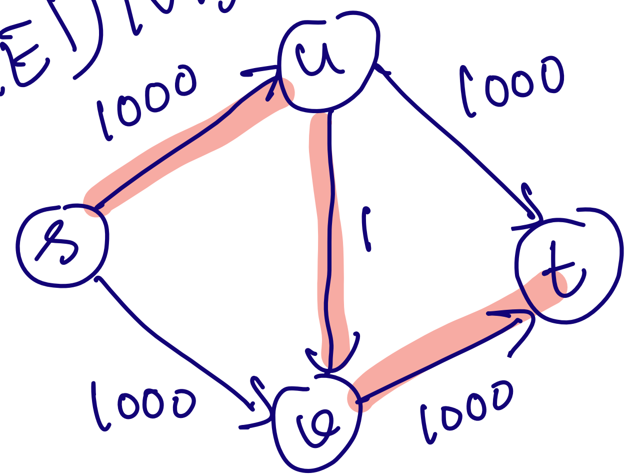


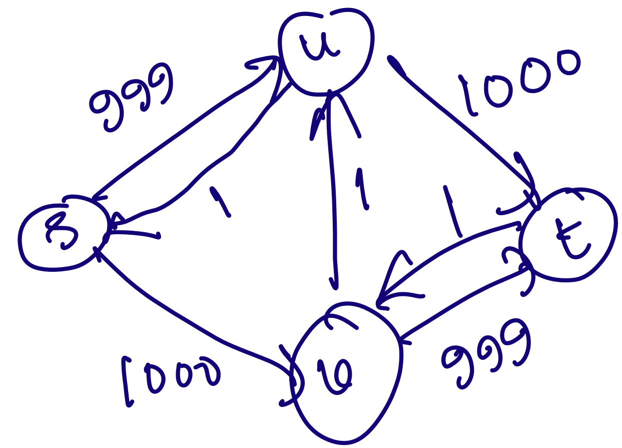


u
s
t
v
20
10
30
10
10
10
10
10
10
30
10
10
10
20
20
10
10
20

- For all $e \in E(G)$, initialize $\mathrm{f}(e) = 0$ and $G_{\mathrm{f}} = G$.
- As long as there is an augmenting path P in G_{f}
    - Send the maximum amount of flow via the path P from s to t. (Find the minimum capacity a^* across all edges of P. Add a to all the edges of P, i.e. for all $e \in P$, $\mathrm{f}(e) \leftarrow \mathrm{f}(e) + a^*$.)
    - Compute G_{f}.
- Output f.

$O(\text{flow value} * (|V| + |E|)|V|)$

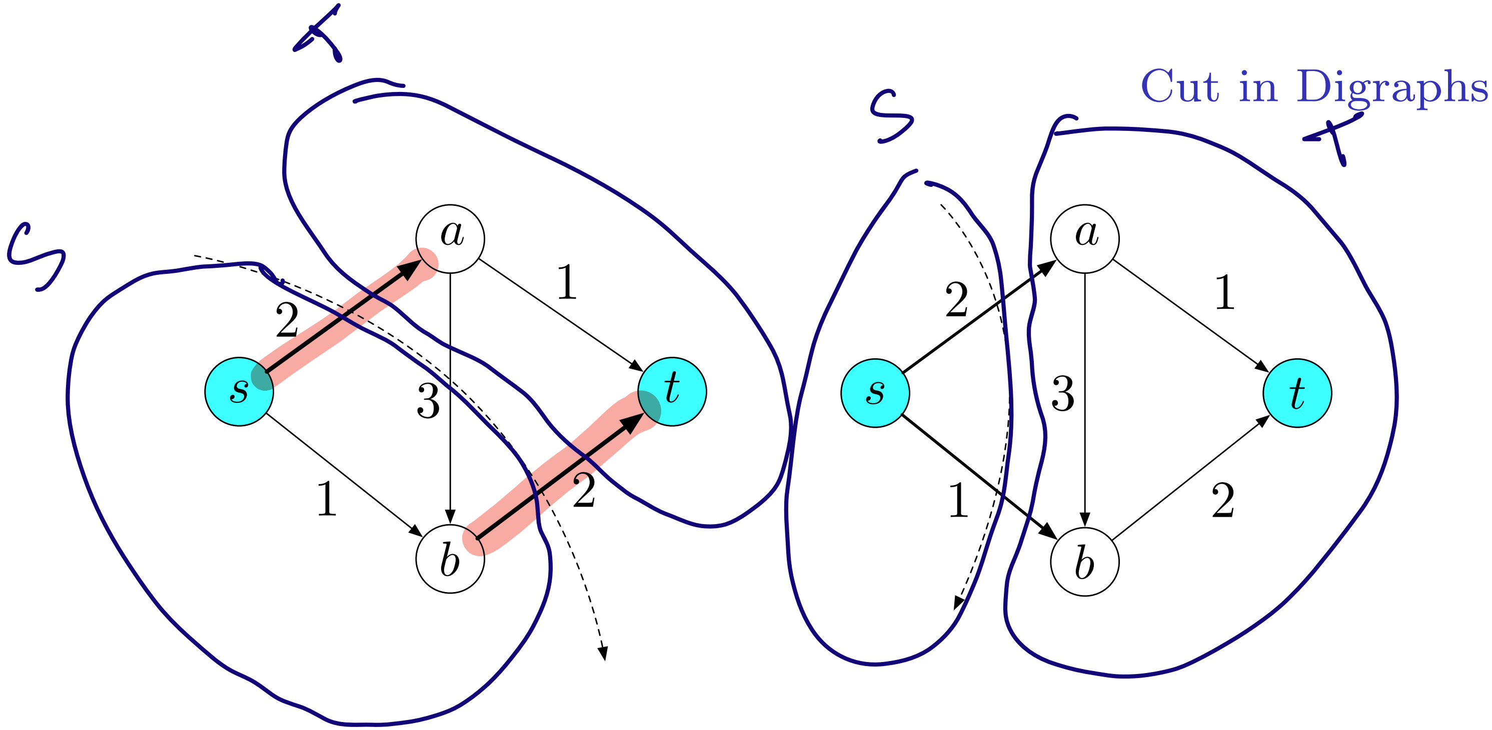




Outline

1. Network Flow
2. Max-Flow Min-Cut Theorem

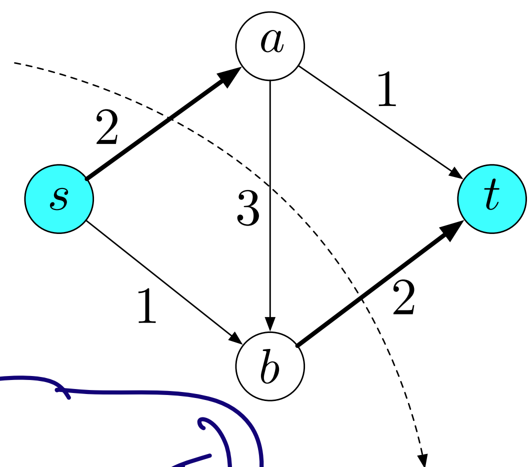
Cut in Digraphs

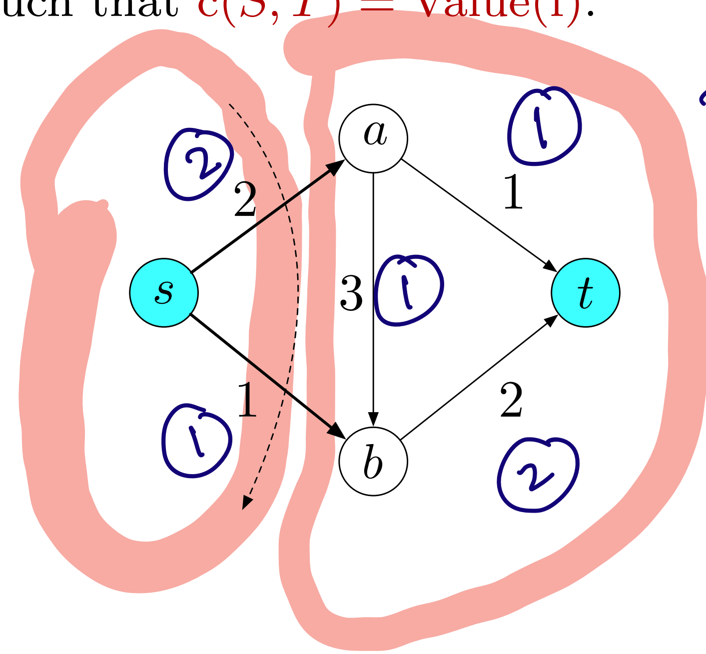


- In a flow network (G, s, t, c), an (s,t)-cut is a nonempty $S \subset V(G)$ such that $s \in S$, and $t \in V \setminus S$.
- Capacity of an (s,t)-cut (S,T) is $\sum_{u \in S} \sum_{v \in T} c(u,v)$. $c(S,T)$
- Capacity of $(\{s,b\},\{a,t\})$ is 4 and capacity of $(\{s\},\{a,b,t\})$ is 3.

Let (G, s, t, c) be a flow network and $f : E(G) \to \mathbb{Z}^+$ be a valid (s,t)-flow. Then, the followings are equivalent.

1. Value(f) is a maximized.
2. G_f has no augmenting paths.
3. There exists an (s,t)-cut (S, T) such that $c(S, T) = \text{Value}(f)$.





flow

$f(s,a) = 2$

$f(s,b) = 1$

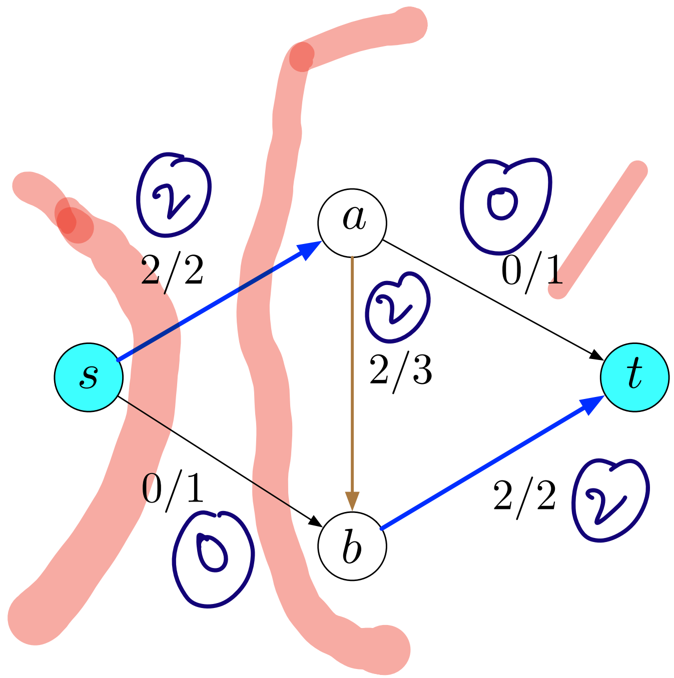
$f(b,t) = 2$

$f(a,t) = 1$

$f(a,b) = 1$

$\text{value}(f) \le c(S,T)$

Flow across a cut

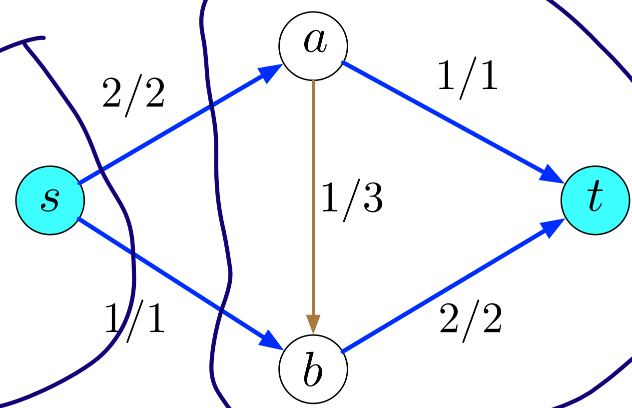


- $\mathrm{f}(S,T) = \sum_{u \in S} \sum_{v \in T} \mathrm{f}(u,v) - \sum_{u \in S} \sum_{v \in T} \mathrm{f}(v,u)$ [flow across (S,T)-cut].

Flow across a cut

S

T



- $f(S,T) = \sum_{u \in S} \sum_{v \in T} f(u,v) - \sum_{v \in T} \sum_{u \in S} f(v,u)$ [flow across the cut (S,T)].
- Flow across $(\{s\},\{a,b,t\})$ is 3.
- Flow across $(\{s,a\},\{b,t\})$ is also 3.

References

- Algorithm Design Chapter 7 by Kleinberg and Tardos.
- Lecture notes by Jeff Erickson.